

1758. Minimum Changes To Make Alternating Binary String

Solved 60%

Easy

Topics

Companies

Hint

You are given a string `s` consisting only of the characters `'0'` and `'1'`. In one operation, you can change any `'0'` to `'1'` or vice versa.

The string is called alternating if no two adjacent characters are equal. For example, the string `"010"` is alternating, while the string `"0100"` is not.

Return *the **minimum** number of operations needed to make `s` alternating.*

Example 1:

Input: `s = "0100"`

Output: `1`

Explanation: If you change the last character to `'1'`, `s` will be `"0101"`, which is alternating.

Example 2:

Input: `s = "10"`

Output: `0`

Explanation: `s` is already alternating.

Example 3:

Input: `s = "1111"`

Output: `2`

Explanation: You need two operations to reach `"0101"` or `"1010"`.

Constraints:

- `1 <= s.length <= 104`
- `s[i]` is either `'0'` or `'1'`.

Python:

class Solution:

def minOperations(self, s: str) -> int:

count, n = 0, len(s)

for i in range(n):

count += (ord(s[i]) ^ i) & 1

return min(count, n - count)

JavaScript:

const minOperations = s => {

let count = 0, n = s.length;

for (let i = 0; i < s.length; i++)

if (s[i] ^ (i & 1)) count++;

return Math.min(count, n - count);

};

Java:

```
class Solution {  
    public int minOperations(String s) {  
        int count = 0, n = s.length();  
        for (int i = 0; i < n; i++)  
            count += (s.charAt(i) ^ i) & 1;  
  
        return Math.min(count, n - count);  
    }  
}
```